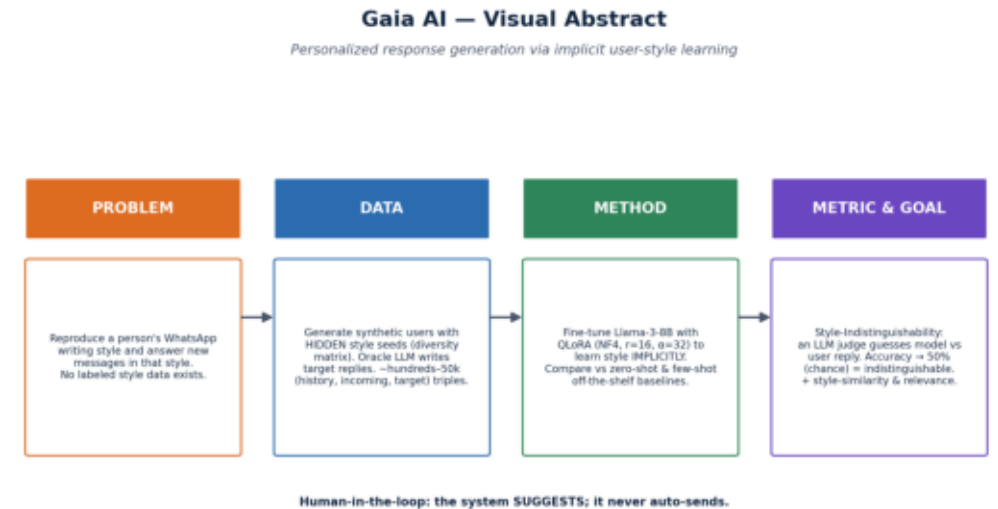


# Gaia AI — Final Report

Personalized response generation via implicit user-style learning · Adir · LLM course (HIT)

# 1 · Project Definition (refined)

- Problem: given a user's chat history + a new message, generate a reply in the user's IMPLICIT style.
- Models: fine-tuned Llama-3-8B (QLoRA) vs off-the-shelf zero-/few-shot; oracle = upper bound.
- Data: synthetic, persona-conditioned, bilingual; per-user splits.
- Metrics: Style-Indistinguishability (headline), style-similarity, relevance.



## 2 · Achievements & Novelty

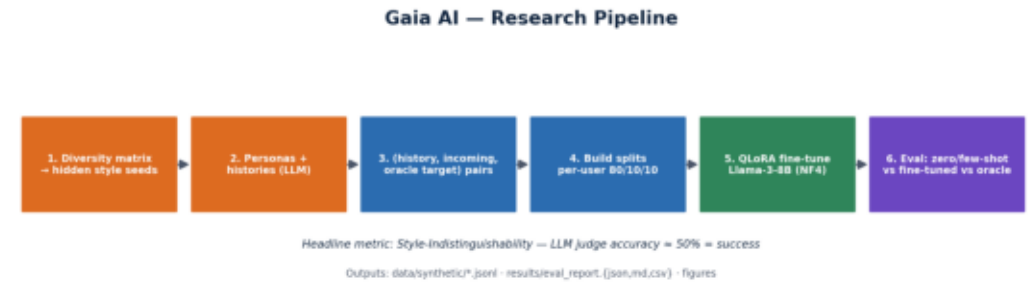
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- Built an end-to-end pipeline: diversity matrix → synthetic data → fine-tune → indistinguishability eval.
- Generated a bilingual persona-conditioned dataset with hidden style seeds (no style labels leaked).
- Produced real baseline results and a reproducible eval (JSON/MD/CSV + figures).
- Novelty: implicit style learning + indistinguishability metric — distinct from retrieval-based personalization (LaMP) and explicit personas (PersonaChat).

# 3 • Methodology

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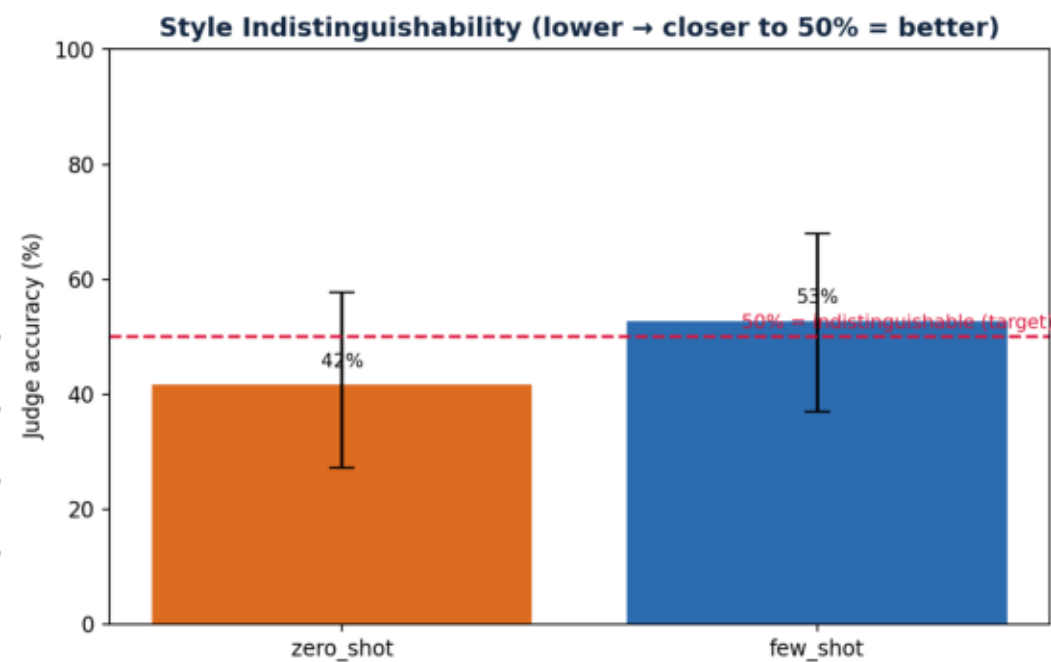
- Synthetic generation: LLM role-play with hidden seeds; oracle writes targets; robust JSON parsing.
- Fine-tuning: QLoRA (4-bit NF4,  $r=16$ ,  $\alpha=32$ , target q/k/v/o) on Llama-3-8B-Instruct (cloud GPU).
- Evaluation effort for quality:
  - Per-user splits; order-randomized judge; judge model separated from generator.
  - Multilingual embedder for he+en style similarity; relevance guards content.
  - Rate-limit-aware generation (backoff, serialized) for reproducibility on free tiers.



## 4 · Results

- Evaluated on a held-out per-user test split (test size = 18).
- Off-the-shelf baselines — zero-shot indist. 42%, few-shot 53% (closer to 50% = better).
- `fine_tuned` column is filled after the QLoRA GPU run (Modal/RunPod) — the core comparison.

| Model      | Indist.       | 95% CI   |
|------------|---------------|----------|
| zero_shot  | 41.7%         | [27,58]% |
| few_shot   | 52.8%         | [37,68]% |
| fine_tuned | pending (GPU) | —        |



*Indistinguishability  $\approx$  50% = success. Figures in visuals/results/.*

## 5 · Conclusion & Future Work

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- The framework works end-to-end and yields real baseline numbers; fine-tuned arm is one GPU run away.
- What I'd change: larger dataset (paid LLM tier), a stronger separate judge model, more error analysis.
- Future: real opt-in data generalization test; RLHF from user edits; multi-personality switching.
- Human-in-the-loop throughout — the system suggests; the user approves.